

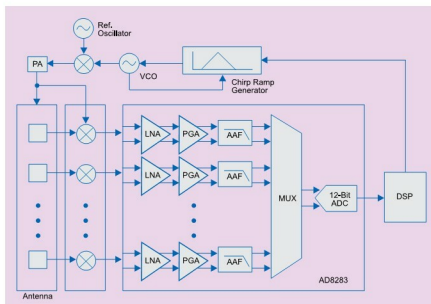


Fig. 6 (a): Radar system overview of AD8283, a six-channel radar receive path AFE

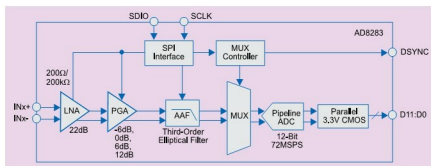


Fig. 6 (b): Simplified block diagram of a single channel of AD8283

provides sufficient resolution.

To fit the radar into the flat bumper assembly and its limited space, it is necessary to use a highly-integrated design, including using part of the radar sub-system PC board as its antenna. Also required are active components such as AD8283 from Analog Devices, which integrates six channels of LNA, a programmable gain amplifier (PGA) and an anti-aliasing filter (AAF) plus one direct-to-ADC channel, with a single 12-bit analogue-to-digital converter [Figs 6 (a) and 6(b)].

Automotive radar systems require sophisticated, controllable analogue front-end circuitry to handle reflected pulse signals across multiple receiver channels, and AD8283 from Analog Devices, which is designed specifi-

cally for this situation.

Primary application for AD8283 is in high-speed ramp, frequency-modulated, continuous-wave radar (HSR-FMCW radar). Performance of each functional block is optimised to meet the demands of this radar system with a careful balance among parameters such as LNA noise, PGA gain range, AAF cutoff characteristics, and ADC sample rate and resolution.

The AD8283 includes a multiplexer in front of the ADC, which automatically switches between each active channel after each ADC sample has been taken. Each channel features a gain range of 16 to 34dB in 6dB increments and an ADC with a conversion rate of up to 72MSPS (mega samples per second).

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Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)
Fax : 0091-22-27673310, 27673330
Email : sales@mecoind.com
Web : www.mecoind.com